

Exercise 1: Maximum

Write a function `my_max(x,y)` that returns the maximum of `x` and `y`. Do not use the `max` function, but use `if` instead in following two ways:

- (a) Use both `if` and `else`.
- (b) Use `if` but not `else` (nor `elif`).

Exercise 2: For loops

Use a `for` loop to:

- (a) Print the numbers 0 to 100
- (b) Print the numbers 0 to 100 that are divisible by 7
- (c) Print the numbers 1 to 100 that are divisible by 5 but not by 3
- (d) Print for each of the numbers $x = 2 \dots 20$, all numbers that divide x , excluding 1 and x . Hence, for 18, it should print 2 3 6 9.

Exercise 3: While loops

Use a `while` loop to find the first 20 numbers that are divisible by 5, 7 and 11, and print them Hint: store the number found so far in a variable.

Exercise 4: Collatz sequence

A Collatz sequence is formed as follows: We start with some number x_0 , and we find the next number in the sequence by

$$x_{i+1} = \begin{cases} \frac{x_i}{2}, & \text{if } x_i \text{ is even} \\ 3x_i + 1, & \text{if } x_i \text{ is odd} \end{cases}$$

If $x_i = 1$, we stop iterating and have found the full sequence.

For example, if we start with $x_0 = 5$, we obtain the sequence:

5 16 8 4 2 1

It is conjectured, though not proven, that every chain eventually ends at 1.

Print the Collatz sequence starting at $x_0 = 103$.

Exercise 5:

Print the numbers that are composed of 4 digits (1,2,3 and 4), and each of the digits is different